

Ramco Institute of Technology
Department of Computer Science and Engineering
Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Civil, EEE, ECE

Course Code & Title: OCS752 Introduction to C Programming

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit 5 / Passing an Entire structure.
 - **Course Outcome:** The student will be able to develop C program using structures.
 - **Topic Learning Outcome:** TLO9.
 - **Activity Chosen:** Think Pair Share
 - **Justification:**
 - After teaching the concept passing a structure to function, I thought of conducting this activity for making the students to understand the concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.
 - **Time Allotted for the Activity:** 20 Minutes
 - **Details of the Implementation:**
 - After teaching the concept of passing a structure to function, the students were made to pair with their neighbours (i.e) In a Class; two students will be seated in a desk.
 - I asked the students to think about passing structure to function concept for 2 minute.
 - Then I told them to Pair with their neighbours and discuss about the answer for their allotted question.
 - Finally, I selected on Pair from each column randomly and ask them to share one their answer for the allotted questions. (3 minutes)
- The sample implementation as shown in figure1,2.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO1	PSO2
CO2	2	2	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO1	PSO2
	2	3	3	3	3	2
Justification for correlation	The basic knowledge of engineering fundamentals is necessary to develop C programs using recursive	Basic C programming concepts using passing an entire structure are used in analysis of complex	The student will be able to develop complex problems using C program		The student can be able to apply C programming using passing an entire structure concepts in emerging trends in Information and	The student can be able to develop C programs using passing an entire structure

	functions	engineering problems	with passing an entire structure		Communication	are used for developing embedded system
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• Images / Screenshot of the practice:

passing a structure into function.

```
#include <stdio.h>
struct date {
    int day;
    char month[10];
    int year;
};
int main() {
    struct date d;
    printf("Enter the day, month and year:");
    scanf("%d %s %d", &d.day, d.month, &d.year);
    display(d);
    return 0;
}

void display(struct date *p) {
    printf("day = %d\n", p->day);
    printf("month = %s\n", p->month);
    printf("year = %d\n", p->year);
}

// Enter the day, month and year : 18 JAN 2022
day = 18
month = JAN
year = 2022
```

Page No.
Date

```
#include <stdio.h>
#include <string.h>
struct Lecturer {
    char Lecturer-Name[50];
    int Total-Experience;
    int Experience-In-This-College;
};
void func(struct Lecturer * Lectures);
int main() {
    struct Lecturer Lecturer1;
    printf("Enter the lecturer name\n");
    scanf("%s", &Lecturer1.Lecturer-Name);
    printf("Enter lecturer Total Years of Experience\n");
    scanf("%d", &Lecturer1.Total-Experience);
    printf("Enter lecturer Total years of Experience in this college\n");
    scanf("%d", &Lecturer1.Experience-In-This-College);
    printf("Lecturer Name = %s, Lecturer1.Lecturer-Name\n");
    printf("Lecturer Total years of Experience = %d, Lecturer1.Total-Experience\n");
    printf("Lecturer Experience in this college = %d, Lecturer1.Experience-In-This-College\n");
    func(&Lecturer1);
}
```

passing structure in function.

```
#include <stdio.h>
#include <string.h>
struct details-of-employee {
    char name[20];
    int id;
    int age;
    int salary;
    int phone-no;
} emp[5];
void func(struct details-of-employee emp);
int main() {
```

```
int i;
printf("Enter the employee details");
for(i=1; i<=3; i++) {
    printf("The detail of Employee %d", i);
    scanf("%d", &emp[i].id);
    scanf("%s", emp[i].name);
    scanf("%d", &emp[i].age);
    scanf("%d", &emp[i].salary);
    scanf("%d", &emp[i].phone-no);
}
func(emp);
return 0;
```

Figure1 Sample Papers

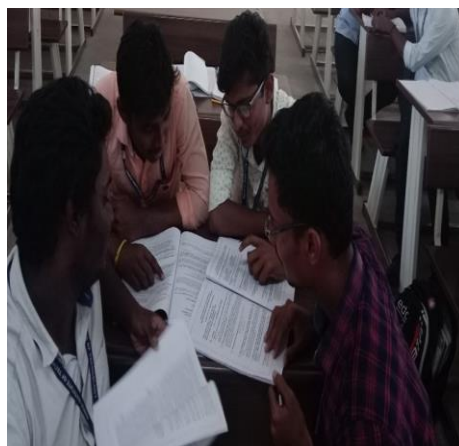


Figure 2 Sample Screen shots

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders***

While using this activity, the students will get clarity in the concept. The students asked to conduct these type of activity in future.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ❖ Students can able to understand passing an entire structure clearly.
- ❖ Communication skills of students improved.

❖ ***Challenges faced in implementation:***

- ❖ Some of the students difficult to understand the concept initially, after discussing with their pair they understood clearly.
- ❖ Some students are hesitating to come in front of a class.
- ❖ The time limit exceeded 10 minutes.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD